

Path Queries II

Time limit: 1.00 s

Memory limit: 512 MB

You are given a tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$. Each node has a value. Your task is to process following types of queries:

1. change the value of node s to x
2. find the maximum value on the path between nodes a and b .

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each query is either of the form " $1\ s\ x$ " or " $2\ a\ b$ ".

Output

Print the answer to each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b, s \leq n$
- $1 \leq v_i, x \leq 10^9$

Example

Input	Output
5 3 2 4 1 3 3 1 2 1 3 2 4 2 5 2 3 5 1 2 2 2 3 5	4 3